

Name: _____



Truncating

Task A: Truncating

1) Complete the table.

Number	Truncate to 1 significant figure	Truncate to 2 significant figures	Truncate to 1 decimal place
89 231.625			
0.05681			
0.0002563			
23.145			
19.99999			

2) Fill in the cross number puzzle.

1				6
2		4		
3				5

Across

2) **Truncate** 63215 to 3 s.f.

3) **Truncate** 0.321986 to 3 d.p.

Down

1) **Truncate** 56842 to thousands.

4) **Truncate** 2331.124 to 3 s.f.

5) Round 13.856 to 2 s.f.

6) Round 10.13986 to 2 d.p.



3) Work out the number given the clues.

a) I am

Truncated to 1 s.f. I am 200

I am a multiple of 12

I am 250 when rounded to the nearest 25

I am a palindrome (number reads the same if all the digits reversed)

b) I am

Truncated to 1 s.f. I am 100

I am a multiple of 4

I am 200 when rounded to the nearest 100

The sum of my digits is 15

Task B: Truncating in real life

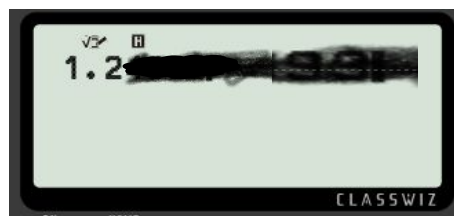
1) Given the scenarios below, identify if the answer should be given exactly, rounded or truncated.

Aisha is 14.7 years old and wants to see a 15 Rated film. Can she go?

A book shop has £89 and wants to buy as many £1.99 puzzle books as possible. How many books can it buy?

A bill of £89 is shared between 10 people.

2) Here is a broken calculator screen.



What is the largest and smallest number that could be displayed on the screen?

3) In computer science, 'INT' or 'int' is used to **truncate** a number so to only give the integers.

For example,

$\text{INT}(56/10)$ will have an output of 5

Work out the output of the following.

- a) $\text{INT}(45/8)$
- b) $\text{INT}(98/10) + \text{INT}(7.8/3)$
- c) $\text{INT}(8/0.5) + \text{INT}(12.65/2) - \text{INT}(9.8/6)$

4) Lucas is going on holiday and looks at the exchange rates.

$\pounds 1 = 1.16887$ Euros

Travel firms and banks offer a truncated value to this:

$\pounds 1 = 1.16$ Euros

- a) Why is truncating exchange rates better for travel firms and banks?
- b) If $\pounds 1\,000\,000$ was exchanged, how much money would the travel companies make due to truncation?